UNIVERSITÄT BIELEFELD
Medical School OWL

MED Bio²
Biostatistics & Medical Biometry

Beyond Residual Plots: Score Plots for Model Diagnosis

Talk at the ZeSt Colloquium
Bielefeld
20.10.2026

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Application Example

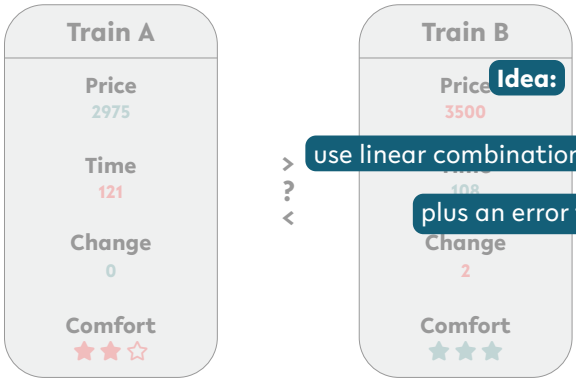
1. Application Example



The *Train* data set as used by Meijer and Rouwendal 2006:

- survey collected in 1987
- **stated choice** between two **two rail alternatives**
- **2929 observations**
- **235 individuals** with **5 to 19 choices**

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Notation

2. Notation

$$
\begin{aligned}
U_{n,j} &= X'_{n,j}\beta^{(X)} + Z'_{n,j}\beta_j^{(Z)} + \varepsilon_{n,j} && n = 1, \ldots, N \\
y_n &= \arg\max_j U_{nj} && j = 1, \ldots, J \\
\Pr(y_n \mid \beta) &= L_n(\beta)
\end{aligned}
$$

$$
\begin{aligned}
ll(\beta) &= \sum_{n=1}^{N} \log(L_n(\beta)) = \sum_{n=1}^{N} ll_n(\beta) \\
\hat{\beta}_{\mathsf{ML}} &= \arg\max_\beta ll(\beta) \\
\iff \partial_\beta ll(\hat{\beta}_{\mathsf{ML}}) &= \sum_{n=1}^{N} \partial_\beta ll_n(\hat{\beta}_{\mathsf{ML}}) = \sum_{n=1}^{N} s_n(\hat{\beta}_{\mathsf{ML}}) \stackrel{!}{=} 0
\end{aligned}
$$

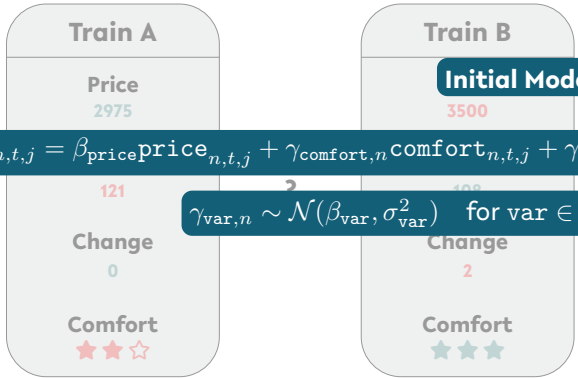
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Initial Model:[1]

$$U_{n,t,j} = \beta_{\text{price}}\text{price}_{n,t,j} + \gamma_{\text{comfort},n}\text{comfort}_{n,t,j} + \gamma_{\text{change},n}\text{change}_{n,t,j} + \gamma_{\text{time},n}\text{time}_{n,t,j} + \varepsilon_{n,t,j}$$

$$\gamma_{\text{var},n} \sim \mathcal{N}(\beta_{\text{var}}, \sigma^2_{\text{var}}) \quad \text{for var} \in \{\text{comfort}, \text{change}, \text{time}\}$$

The *Train* data set as used by Meijer and Rouwendal 2006:

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[1] Same utility formulation and mixed effects as used in Meijer and Rouwendal 2006

Modelling Workflow

3. Modelling Workflow

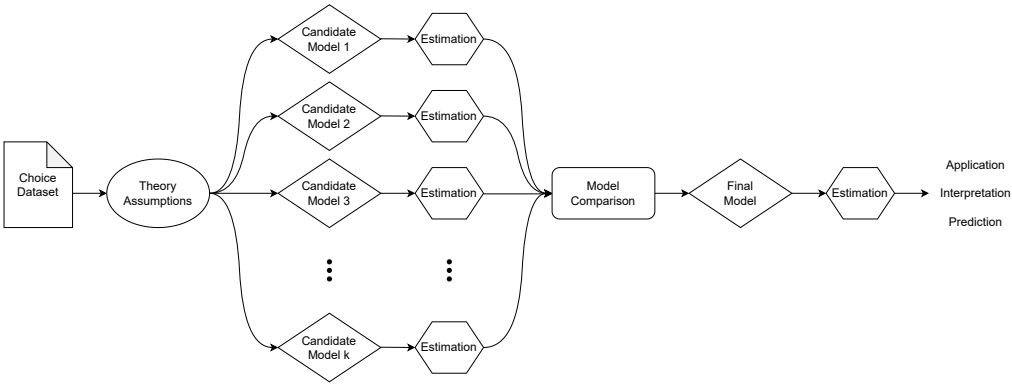


Figure: Adapted from Nova et al. 2024

Model Diagnosis

4. Model Diagnosis

Model diagnosis in **Classical Linear Regression Modelling**:

- **visual model diagnosis** for **non-linear covariate dependencies** ⟶ **residual plots**
- tests for non-linear dependencies ⟶ **Ramsey RESET test**
- **visual model diagnosis** for **heteroscedasticity** ⟶ **residual plots**
- **visual model diagnosis** for **independence of the errors** ⟶ **Standardized residual plots**
- tests for independence of the errors ⟶ **Durbin–Watson test**
- **outlier detection** ⟶ **Studentized residuals** and **Cook's distance**

Residuals are just observational level gradient contributions for the intercept parameter

Gradient usage in DCM

5. Gradient usage in DCM

Gradient information usage in **Choice Modelling**:

- **score function: estimation** of maximum likelihood estimators (score ≈ 0)
- **individual level gradients:** optimization procedures like the **BGW**[2]
- **individual level gradients:** robust **sandwich estimator** for the estimator variance[3]
- **individual level gradients:** inform modelling of **deterministic heterogeneity**, improve **latent class structures** and guide **distribution choice** for continuous mixture models[4]
- **observational-pair level gradients:** Lagrange multiplier type tests for CML estimation to test for intra-individual heterogeneity[5]

[2] Bunch et al. 1993
[3] Qin and Lawless 1994
[4] Hess 2026
[5] Büscher and Bauer 2025

Lemma

Define $s_n^{(k)}(\beta; X_n, y_n) = \partial_{\beta^{(k)}} ll_n(\beta)$. Let β_0 be the true parameter vector governing the DGP, and let $\hat{\beta}$ be an estimator of β_0 such that $P_{n,i}(\hat{\beta})$ is a consistent estimator of $P_{n,i}(\beta_0)$. Furthermore, assume the probabilities for all observed choice occasions n and all choice alternatives i to be bounded from below ($P_{n,i}(\beta) > \delta \; \forall n, i$ for some $\delta > 0$) and that the gradients of the probabilities w.r.t. β are bounded ($\left|\partial_{\beta^{(k)}} P_{n,i}(\beta)\right| < M$ for some $0 < M < \infty$). Then the score contributions conditional expected value will converge to zero as the number of observations N goes to infinity, so

$$\mathrm{E}\left(s_n^{(k)}(\hat{\beta}; X_n, y_n) \mid X_n\right) \overset{N\to\infty}{\longrightarrow} 0.$$

Proof

Sebastian Büscher (2024b). “Visual Guidance for Model Specification: Introducing Score Plots for Discrete Choice Models”. In: *Available at SSRN*. SSRN: 5065076

Lemma

Core Message: Average score information should be zero independent of the covariate values X

Def… let $\hat{\beta}$ be an estimator… thermore, assume the probabilities… tives i to be bounded from below ($P_{n,i}(\beta) > \delta \; \forall n, i$ for some $\delta > 0$) and that the gradients of the probabilities

Idea: Visualise gradient contributions depending on covariate values to detect model misspecifications

$$\mathrm{E}\left(s_n^*(\beta; X_n, y_n) \mid X_n\right) \longrightarrow 0.$$

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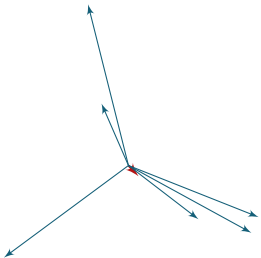
Visualisation

6. Visualisation

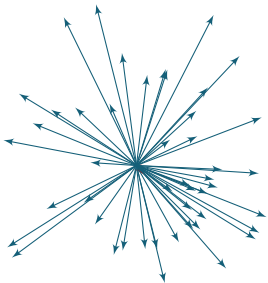
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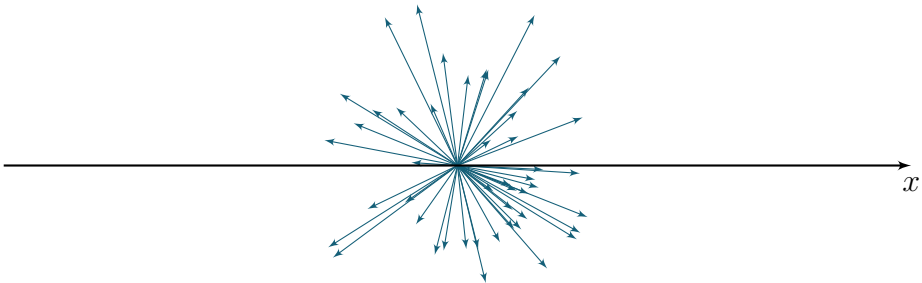
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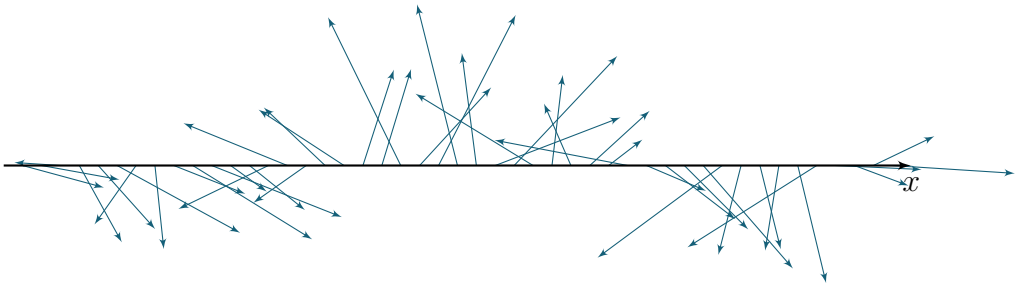
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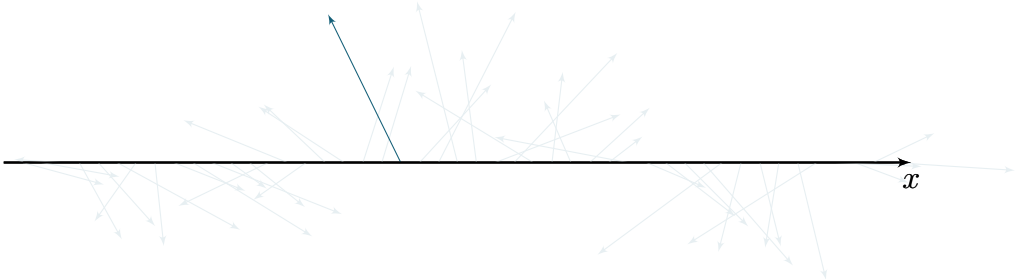
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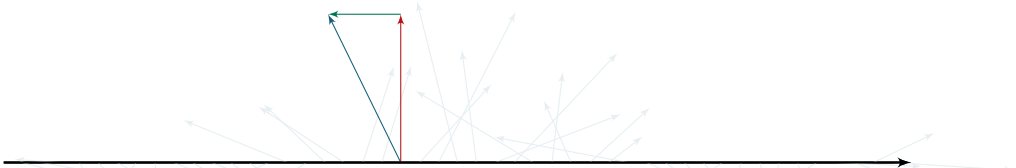
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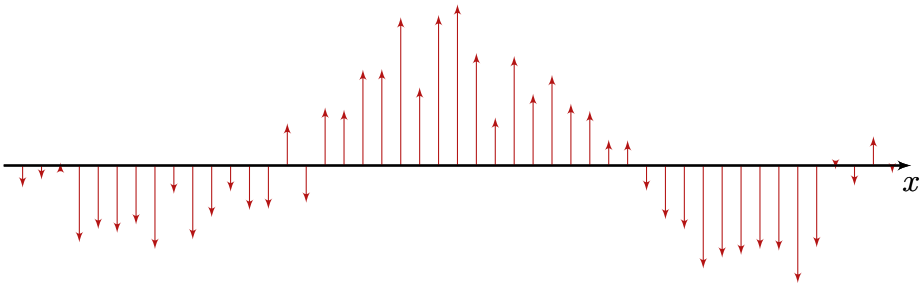
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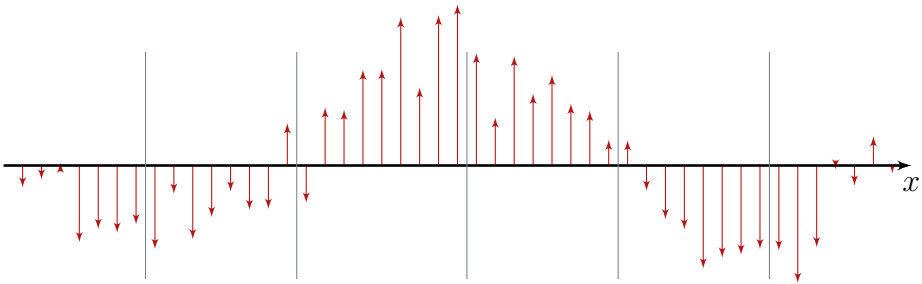


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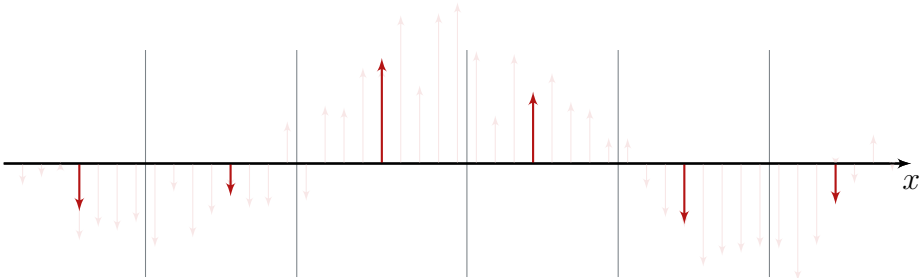


Shiny

6. Visualisation



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6. Visualisation

$s_n(\hat{\beta}) := \partial_\beta ll_n(\hat{\beta})$

score contribution

6. Visualisation

general stats

$$s_n(\hat{\beta}) := \partial_\beta ll_n(\hat{\beta})$$ score contribution

$$\overline{s^{(k)}}(\theta; X, y) := \frac{\sum_n s_n^{(k)}(\theta; X_n, y_n)}{N}$$ mean score contribution

$$\widehat{\mathrm{sd}}(s^{(k)}(\theta; X, y)) := \sqrt{\frac{\sum_n \left(s_n^{(k)}(\theta; X_n, y_n) - \overline{s^{(k)}}(\theta; X, y)\right)^2}{N-1}}$$ emp. std. dev. of scores

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local mean

$$\Pi_p(x) := \begin{cases} 1 & x \in [\kappa_{p-1}, \kappa_p) \\ 0 & \text{else} \end{cases}$$ $\kappa_0, \ldots, \kappa_P \in \mathbb{R}$

$$\overline{s_p^{(k)}}(\theta; X, y) := \frac{\sum_n \Pi_p(x_n) s_n^{(k)}(\theta; X_n, y_n)}{\sum_n \Pi_p(x_n)}$$ local mean score

$$\overline{\text{rel. score}}_p^{(k)}(\theta) := \frac{\overline{s_p^{(k)}}(\theta; X, y) - \overline{s^{(k)}}(\theta; X, y)}{\widehat{\mathrm{sd}}(s^{(k)}(\theta; X, y))}$$ **std. local mean score**

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mean local deviance / sd

$$\widehat{\mathrm{sd}}(s_p^{(k)}(\theta; X, y)) := \sqrt{\frac{\sum_n \left(\Pi_p(x_n) s_n^{(k)}(\theta; X_n, y_n) - \overline{s_p^{(k)}}(\theta; X, y)\right)^2}{(\sum_n \Pi_p(x_n)) - 1}}$$ local emp. std. dev. of scores

$$\text{score stat}_p^{(k)}(\theta) := \frac{\overline{s_p^{(k)}}(\theta; X, y) - \overline{s_p^{(k)}}(\theta; X, y)}{\widehat{\mathrm{sd}}(s_p^{(k)}(\theta; X, y))} \sqrt{\sum_n \Pi_p(x_n)}$$ **score statistic**

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colour and CIs

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Shiny

Application Example

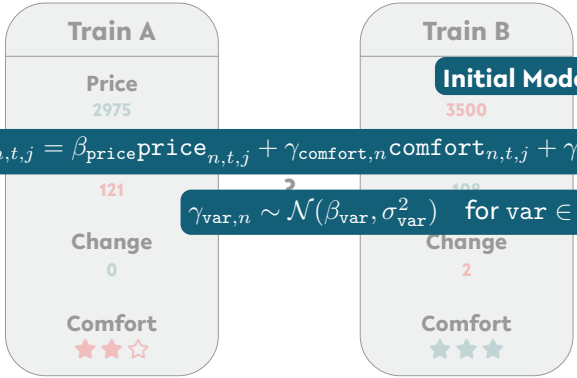
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The *Train* data set:

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Initial Model:[6]

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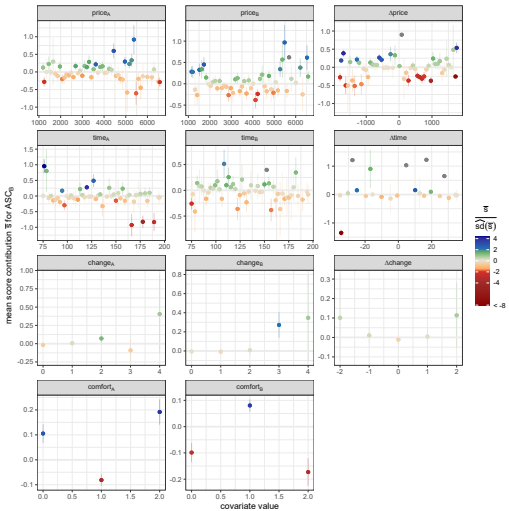
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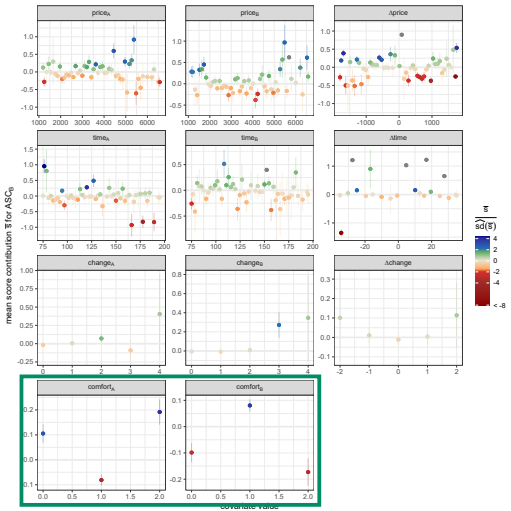
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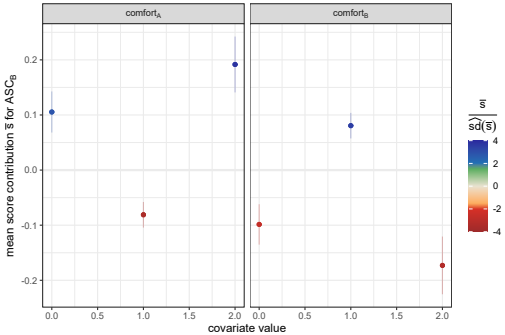
model	ll	AIC	BIC
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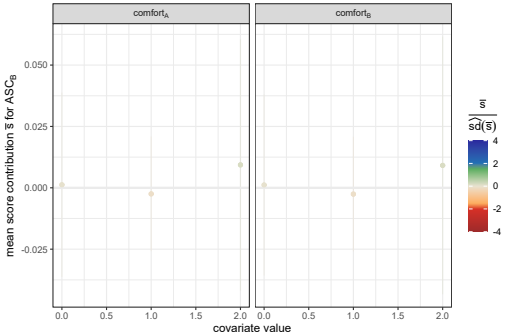
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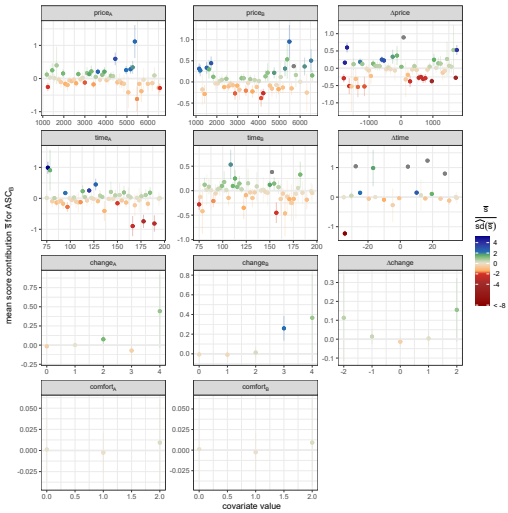
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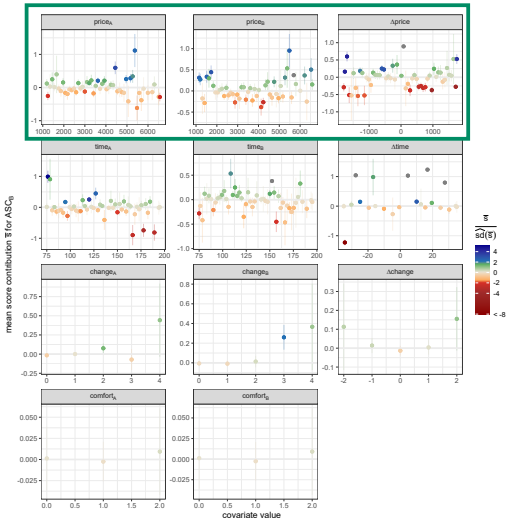
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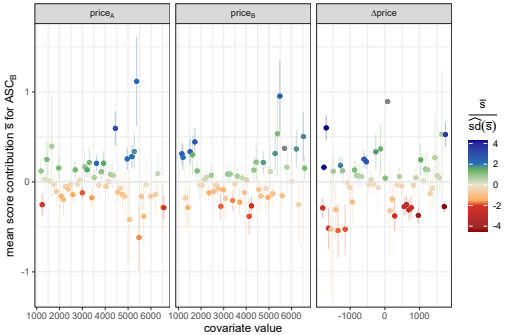
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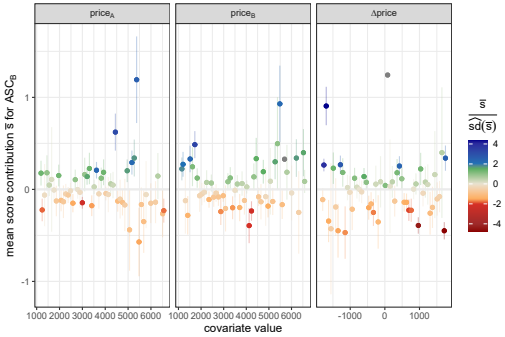
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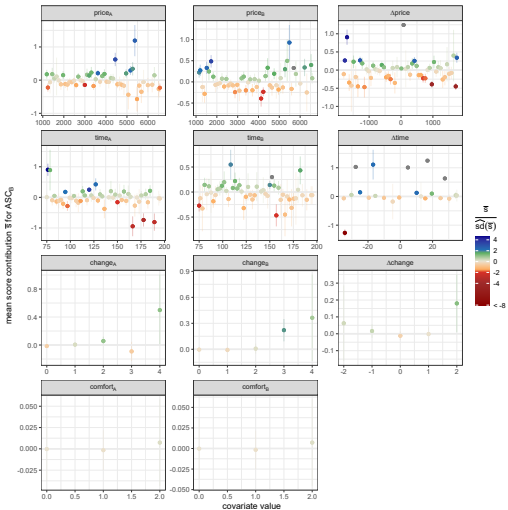
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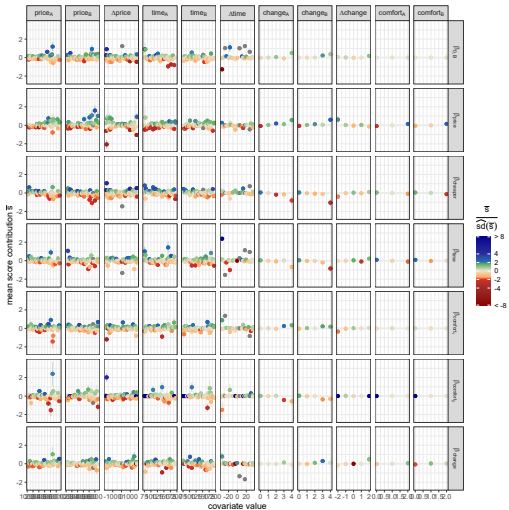
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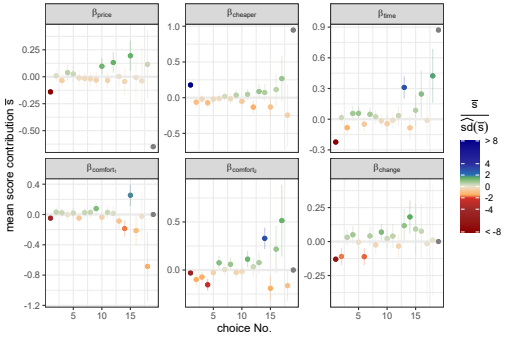
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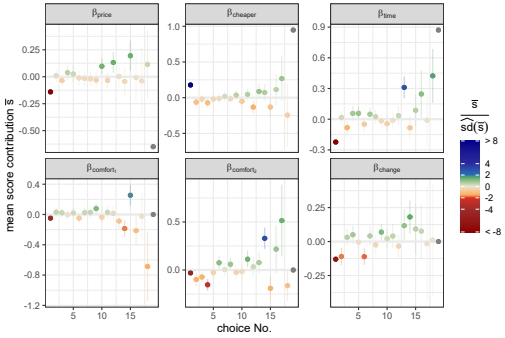
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+ as.factor(comfort)	$-1\,693.78$	$3\,399.55$	$3\,435.45$
+ $I(\Delta\text{price} > 0)$	$-1\,668.41$	$3\,350.81$	$3\,392.69$

7. Application Example



model	ll	AIC	BIC
initial model	$-1\,540.21$	$3\,094.43$	$3\,136.30$
- mixed effects + ASC	$-1\,723.84$	$3\,457.67$	$3\,487.59$
+ as.factor(`comfort`)	$-1\,693.78$	$3\,399.55$	$3\,435.45$
+ $I(\Delta\texttt{price} > 0)$	$-1\,668.41$	$3\,350.81$	$3\,392.69$

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+ $I(\Delta\texttt{price} > 0)$	$-1\,668.41$	$3\,350.81$	$3\,392.69$
+ mixed effects - ASC	$-1\,488.15$	$2\,996.31$	$3\,056.13$
Δ	52.06	-98.12	-80.17

New Modelling Workflow

8. New Modelling Workflow

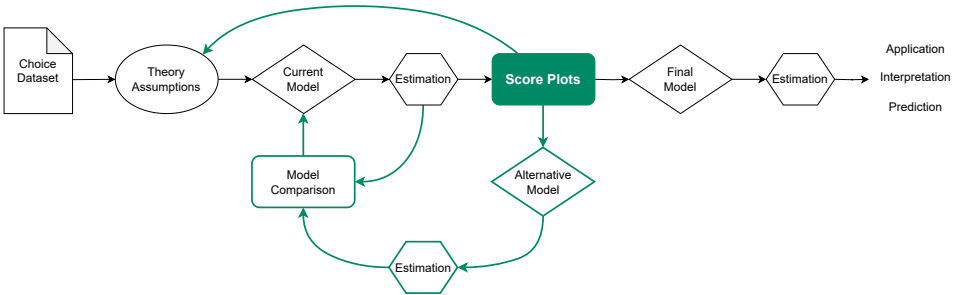


Figure: Revised workflow from Büscher 2024a

Mixed Models

Approaches for mixed models include

- use individual level gradients with mean covariate values
- investigate functional form of the utility function without mixing, add mixed effects for final model
- utilise **phantom gradients**: $\tilde{s}_{-n}^{(k)}(\hat{\beta}) = s^{(k)}(\hat{\beta}) - s_{-n}^{(k)}(\hat{\beta})$
  - $s_{-n}^{(k)}(\hat{\beta})$ is the gradient off all observations except of observation n
  - requires more computation time
  - not equivalent to gradient contributions
  - do not necessarily add up to zero

Outlook

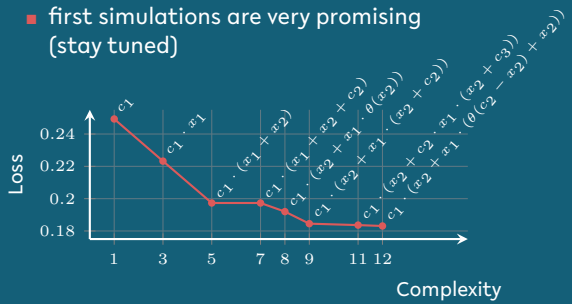
10. Outlook

R package:

- making visual gradient diagnostics easily accessible to practitioners
- extend to wider range of models (`glm`, `mlogit`, `Rprobit`, ...)
- extend to mixed models
- develop continuous estimation of expected conditional gradient contribution (nearest neighbour estimator)

Symbolic Regression:

- collaboration with **Dr. Kaja Balzereit**
- learn **functional form of the utility function** from **observational level gradients**
- first simulations are very promising (stay tuned)



$$U = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \beta_3 \cdot x_1 \cdot x_2 + \varepsilon$$

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Appendix

Proof of Lemma.

Using

(a) that the sum of the choice probabilities over all alternatives has to equal one and thus the derivative of the sum is equal to zero ($\sum_{i=1}^{J} \partial_{\beta^{(k)}} P_{n,i}(\hat{\beta}) = 0$) and

(b) that the expected value of the score is the sum of the conditional expectations of the score, conditional on the chosen alternative, times their choice probabilities,

(c) the dominated convergence theorem[7], which, applied to sequence of random variables $\{X_n\}$ states that if (1) $\exists X$ such that $X_n \xrightarrow{P} X$ and (2) $\exists Y$ with $\mathrm{E}(|Y|) < \infty$ such that $|X_n| \leq Y \; \forall n \in \mathbb{N}$, then $\mathrm{E}(X_n) \to \mathrm{E}(X)$, and

(d) $\sum_{i=1}^{J} \frac{\partial_{\beta^{(k)}} P_{n,i}(\hat{\beta})}{P_{n,i}(\hat{\beta})} P_{n,i}(\beta_0) < \sum_{i=1}^{J} \frac{M}{\delta} P_{n,i}(\beta_0) = \frac{M}{\delta} < \infty$,

we get for a consistent estimator of $P_{n,i}(\beta_0)$:

Slides

[7] Çinlar 2011

Proof of Lemma (continuation).

$$\begin{aligned}
\lim_{N\to\infty} \mathrm{E}\left(s_n^{(k)}(\hat{\beta}; X_n, y_n) \mid X_n\right) &= \lim_{N\to\infty} \mathrm{E}\left(\sum_{i=1}^{J} \frac{\partial_{\beta^{(k)}} P_{n,i}(\hat{\beta})}{P_{n,i}(\hat{\beta})} P_{n,i}(\beta_0) \mid X_n\right) \\
&= \mathrm{E}\left(\lim_{N\to\infty} \sum_{i=1}^{J} \partial_{\beta^{(k)}} P_{n,i}(\hat{\beta}) \underbrace{\frac{P_{n,i}(\beta_0)}{P_{n,i}(\hat{\beta})}}_{\overset{N\to\infty}{\longrightarrow} 1} \mid X_n\right) \\
&= \mathrm{E}\left(\lim_{N\to\infty} \underbrace{\sum_{i=1}^{J} \partial_{\beta^{(k)}} P_{n,i}(\hat{\beta})}_{=0} \mid X_n\right) \\
&= 0.
\end{aligned}$$

□

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